

STAREHE GIRLS' 2025 PREDICTION



POSSIBLE KCSE EXAM

233/3

CHEMISTRY

PAPER 3 (PRACTICAL)

TIME: 2¼ HOURS



NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES

- ❖ Write your name, index number, name of the school and the date in the spaces provided.
- ❖ You are required to spend 15 minutes of the 2¼hrs reading through the paper and make sure you have all the apparatus and chemicals needed for the practical.
- ❖ Answer all the questions in the spaces provided after each question
- ❖ Electronic calculators and mathematical tables may be used
- ❖ All working must be clearly shown where necessary.

For Examiners Use Only

Question	Maximum score	Candidate's Score
1	23	
2	10	
3	07	
Total	40	

1) You are provided with the following

- An acid labeled solution H
- 2.0M sodium hydroxide solution labelled solution J
- Solution K containing 25 g per litre of an alkanolic acid

You are required to

- Prepare a dilute solution of sodium hydroxide solution J
- Determine the:
 - i) Molar mass of alkanolic acid, solution K.
 - ii) Reaction ratio between sodium hydroxide and acid H

Procedure 1:

Using a pipette and pipette filler, place 25 cm³ of solution J into a 250 ml volumetric flask. Add about 150 cm³ of distilled water and shake well. Add more distilled water to make up to the mark. Label this as solution L. **Retain the remaining solution J for use in procedure 2.**

Fill the burette with solution K. Using clean pipette and pipette filler, place 25 cm³ of solution L in to a 250 ml conical flask. Add 2 drops of phenolphthalein indicator and titrate with solution K. Record your results in the table I. Repeat the titration two more times and complete the table.

Table I

(4 marks)

	I	II	III
Final Burette Reading (cm ³)			
Initial Burette Reading (cm ³)			
Volume Of Solution K (cm ³)			

a) Calculate:

i) The average volume of solution K used.

(1 mark)

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ii) Concentration of solution L in moles per litre.

(1mark)

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iii) Concentration of the alkanoic acid solution K in moles per litre. (1 mole of the acid reacts with 3 moles of the base.) (1mark)

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iv) Molar mass of the alkanoic acid (1mark)

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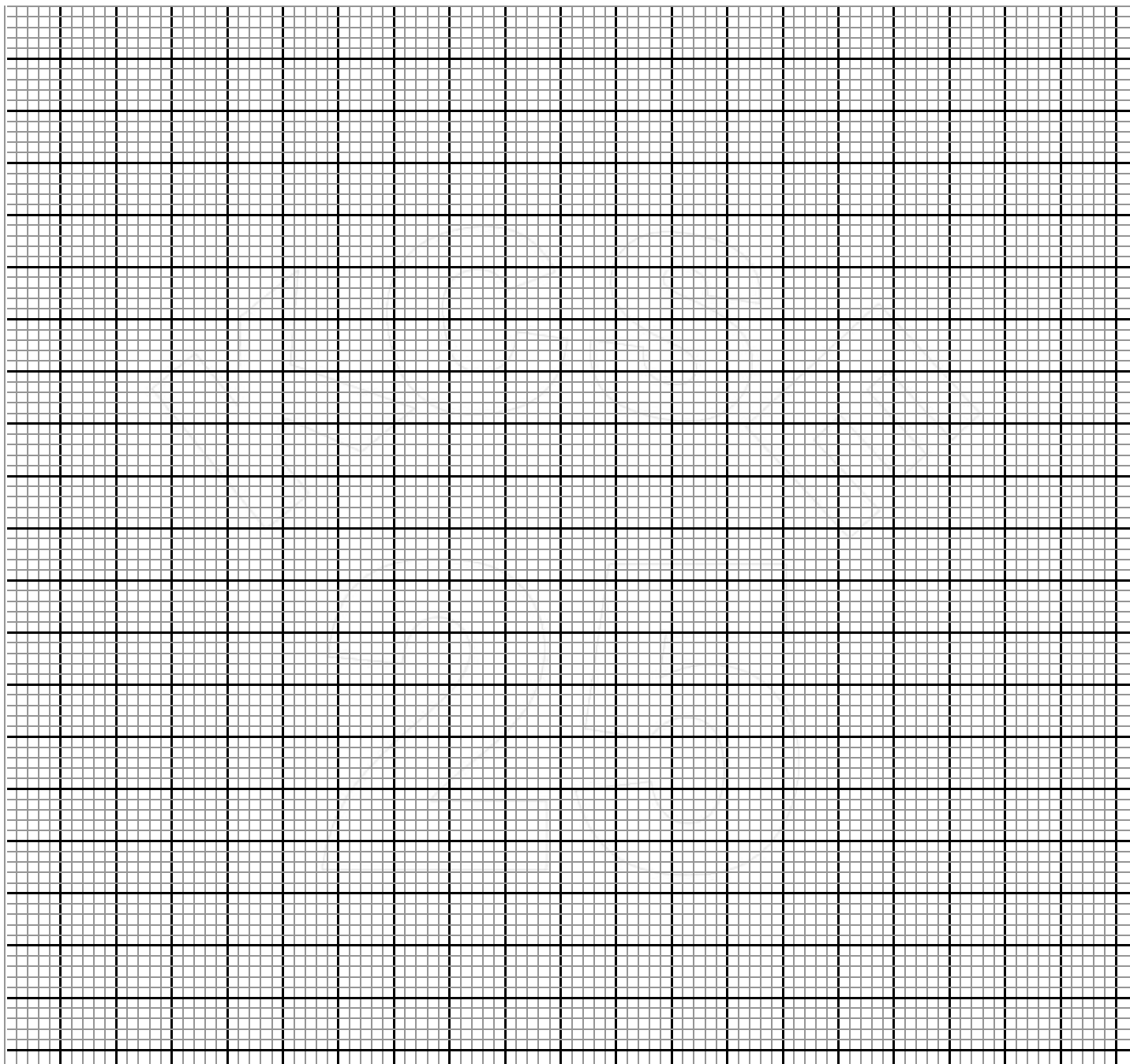
PROCEDURE 2

Fill a clean burette with solution H. Place 5 cm³ of solution H in 100 ml plastic beaker. Measure the initial temperature of solution H in the beaker and record it in the table II. Using a 50 ml measuring cylinder, measure 25 ml of solution J. add it to solution H in the beaker and immediately stir the mixture with the thermometer. Record the maximum temperature reached in table II. Repeat the experiment with other sets of volumes of solutions H and J and complete the table 2 below

Table 2. (6 marks)

Volume of solution H(cm ³)	5	9	13	17	21	25
Volume of solution J (cm ³)	25	21	17	13	9	5
Maximum temperature (°C)						
Initial temperature (°C)						
Change in temperature (°C)						

- a) On the grid provided, plot a graph of **change in temperature** against the **volume of solution H**.
(3 marks)



- b) From the graph, determine the volume of solution H that was needed for complete neutralization.
(1 mark)

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c) Calculate the volume of solution J that fully reacted with the solution H stated in (b) above.

(1 mark)

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d) Calculate the:

i) Ratio between the volumes of solutions H and J that neutralized one another. **(1 mark)**

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ii) Concentration in moles per litre of the acid in solution H. (Assume that the volume ratio is the same as mole ratio). **(1 mark)**

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e) Calculate the molar heat of neutralization of sodium hydroxide solution in kJmol^{-1} given that the specific heat capacity for water is 4.2J/g/K and the density for water is 1g/cm^3 . **(2marks)**

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a) Place all of the solid P in a boiling tube. Add about 10 cm³ of distilled water and shake well.

Divide the mixture into **five portions**.

b) To the first portion of solution in a test tube, add 2 - 3 drops of aqueous potassium iodide solution.

Observation	Inferences
1mark	1 mark

Observation	inferences
1 mark	1 mark

d) To the third portion of the solution in a test tube, add 2M ammonia solution drop wise until in excess.

Observation	inferences
1 mark	$\frac{1}{2}$ mark

e) To the fourth portion of the solution in a test tube, add 2 - 3 drops of barium nitrate solution.

Observation	inferences
$\frac{1}{2}$ mark	1 mark

f) To fifth portion of the solution in a test tube, add 2 - 3 drops of lead (II) nitrate solution.

Observation	inferences
$\frac{1}{2}$ mark	$\frac{1}{2}$ mark

3) You are provided with substance U. You are required to

- Carry out tests on the substance U

-Record all observations and inferences accordingly.

(i) Scoop half a spatula end full of substance U and ignite it using a non-luminous flame.

Observation	Inferences
$\frac{1}{2}$ mark	$\frac{1}{2}$ mark

(ii) Place the remaining substance U in a boiling tube, add about 8 cm³ distilled water and shake the mixture well. Divide the mixture into three portions.

Observation	Inferences
1 mark	$\frac{1}{2}$ mark

(iii) To the first portion in a test-tube, Add $\frac{1}{2}$ spatula end full of solid sodium hydrogen carbonate.

Observation	Inferences
$\frac{1}{2}$ mark	$\frac{1}{2}$ mark

